List of Listings

Chapter 1. Hello, world of concurrency in C++!

Listing 1.1. A simple Hello, Concurrent World program

Chapter 2. Managing threads

Listing 2.1. A function that returns while a thread still has access to local variables

Listing 2.2. Waiting for a thread to finish

Listing 2.3. Using RAII to wait for a thread to complete

Listing 2.4. Detaching a thread to handle other documents

Listing 2.5. Returning a std::thread from a function

Listing 2.6. scoped_thread and example usage

Listing 2.7. Spawn some threads and wait for them to finish

Listing 2.8. A naïve parallel version of std::accumulate

Chapter 3. Sharing data between threads

Listing 3.1. Protecting a list with a mutex

Listing 3.2. Accidentally passing out a reference to protected data

Listing 3.3. The interface to the std::stack container adapter

Listing 3.4. An outline class definition for a thread-safe stack

Listing 3.5. A fleshed-out class definition for a thread-safe stack

Listing 3.6. Using std::lock() and std::lock_guard in a swap operation

Listing 3.7. Using a lock hierarchy to prevent deadlock

Listing 3.8. A simple hierarchical mutex

Listing 3.9. Using std::lock() and std::unique_lock in a swap operation

Listing 3.10. Locking one mutex at a time in a comparison operator